

Michael Xavier Canonizado
BSCS 3A
Data Mining
Apriori Algorithm Task

Dataset

TID	Items
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Bread, Butter

Task 1: Find Frequent Itemsets (L1 & L2)

- 1) Compute support for all 1-itemsets
- 2) Compute support for all 2-itemsets
- 3) Keep only those with support ≥ 2

L1

Items (L1)	Count	Support
Milk	3	$\frac{3}{5} = 0.6$
Bread	4	$\frac{4}{5} = 0.8$
Butter	4	$\frac{4}{5} = 0.8$

L2

Items (L2)	Count	Support
Milk, Bread	2	$\frac{2}{5} = 0.4$
Milk, Butter	2	$\frac{2}{5} = 0.4$
Bread, Butter	3	$\frac{3}{5} = 0.6$

Task 2: Generate ALL Possible Rules

Task 3: Compute Metrics (Support, Confidence, Lift)

Task 4: Filter Strong Rules

- Support ≥ 0.4
- Confidence ≥ 0.6
- Lift > 1

Rule	Support	Confidence	Lift	Interpretation
{ Milk , Bread }				
Milk $\rightarrow$ Bread	$\frac{2}{5} = 0.4$	$\frac{2}{3} = 0.67$	$0.67 / 0.8 = 0.8375$	Don't Keep. Low lift
Bread $\rightarrow$ Milk	$\frac{2}{5} = 0.4$	$2/4 = 0.5$	$0.5 / 0.6 = 0.8333$	Don't Keep. Low lift & confidence
{ Milk , Butter }				
Milk $\rightarrow$ Butter	$\frac{2}{5} = 0.4$	$\frac{2}{3} = 0.67$	$0.67 / 0.8 = 0.8375$	Don't Keep. Low lift
Butter $\rightarrow$ Milk	$\frac{2}{5} = 0.4$	$2/4 = 0.5$	$0.5 / 0.6 = 0.8333$	Don't Keep. Low lift & confidence
{ Bread , Butter }				
Bread $\rightarrow$ Butter	$\frac{3}{5} = 0.6$	$\frac{3}{4} = 0.75$	$0.75 / 0.8 = 0.9375$	Don't Keep. Low lift
Butter $\rightarrow$ Bread	$\frac{3}{5} = 0.6$	$\frac{3}{4} = 0.75$	$0.75 / 0.8 = 0.9375$	Don't Keep. Low lift